

Calibration under truncation: working notes and proposed direction

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1 Problem statement

While using DagFlow, the user (with assistance from the LLM) declares a target mean and target variance for all variables, as well as a target R^2 for endogenous variables relative to their parents. Each variable is also given declared bounds. Under the current implementation, this truncation applied during sampling isn't accounted for in the calibration equations. This results in shifts in the realized mean from its declared target, and attenuation of the realized variance and R^2 below their targets.

The goal is to create a calibrator that accounts for the known bounds in order to match the realized mean, variance, and R^2 to the declared targets.

Going forward, we will use the following symbols for consistency:

μ_p = Exogenous parameter mean
 σ_p^2 = Exogenous parameter variance
 μ_t = Target mean
 σ_t^2 = Target variance
 η = Conditional linear predictor, $\beta_0 + \beta_1 X$
 σ_ε^2 = Endogenous error variance
 R_t^2 = Target R^2

Note: For simplicity, we work the single parent case in Sections 3-4. The extension to multiple parents is a planned next step.

2 Exogenous moment calibration

The mean and variance of a $\text{TruncNormal}(\mu, \sigma^2, a, b)$ draw with location parameter μ , variance parameter σ^2 , lower bound a , and upper bound b are given by the functions

$$m(\mu, \sigma) = \mu + \sigma \frac{\varphi(\alpha) - \varphi(\beta)}{Z}, \quad (1)$$

$$v(\mu, \sigma) = \sigma^2 \left[1 + \frac{\alpha \varphi(\alpha) - \beta \varphi(\beta)}{Z} - \left(\frac{\varphi(\alpha) - \varphi(\beta)}{Z} \right)^2 \right], \quad (2)$$

with standardized bounds and surviving mass

$$\alpha = \frac{a - \mu}{\sigma}, \quad \beta = \frac{b - \mu}{\sigma}, \quad Z = \Phi(\beta) - \Phi(\alpha). \quad (3)$$

In DagFlow, an exogenous variable is drawn as $Y \sim \text{TruncNormal}(\mu_p, \sigma_p^2, a, b)$ with μ_p and σ_p^2 parameters of the parent distribution. The realized moments of the draw are

$$\mathbb{E}[Y] = m(\mu_p, \sigma_p) = \mu_p + \sigma_p \frac{\varphi(\alpha) - \varphi(\beta)}{Z}, \quad (4)$$

$$\text{Var}(Y) = v(\mu_p, \sigma_p) = \sigma_p^2 \left[1 + \frac{\alpha \varphi(\alpha) - \beta \varphi(\beta)}{Z} - \left(\frac{\varphi(\alpha) - \varphi(\beta)}{Z} \right)^2 \right], \quad (5)$$

with α, β, Z evaluated at (μ_p, σ_p) giving $\alpha = (a - \mu_p)/\sigma_p$, $\beta = (b - \mu_p)/\sigma_p$.

So, for known target moments μ_t and σ_t^2 , we set

$$m(\mu_p, \sigma_p) = \mu_t \quad \text{and} \quad v(\mu_p, \sigma_p) = \sigma_t^2. \quad (6)$$

We can then solve the system for μ_p and σ_p . Drawing then with these new parameters allows us to hit our target realized moments.

3 Endogenous moment calibration (single parent)

An endogenous variable in DagFlow is drawn conditional on its parent, $Y \mid X \sim \text{TruncNormal}(\eta, \sigma_\varepsilon^2, a, b)$ with $\eta_i = \beta_0 + \beta_1 X_i$. Here, η and σ_ε^2 are parameters of the conditional distribution. The parent is already sampled, so the η_i are computable for any candidate parameters $(\beta_0, \beta_1, \sigma_\varepsilon)$.

With standardized bounds and surviving mass now functions of the conditional mean,

$$\alpha(\eta) = \frac{a - \eta}{\sigma_\varepsilon}, \quad \beta(\eta) = \frac{b - \eta}{\sigma_\varepsilon}, \quad Z(\eta) = \Phi(\beta(\eta)) - \Phi(\alpha(\eta)), \quad (7)$$

the conditional moments are the Section 2 formulas evaluated along the regression:

$$\mathbb{E}[Y \mid X] = m(\eta, \sigma_\varepsilon) = \eta + \sigma_\varepsilon \frac{\varphi(\alpha(\eta)) - \varphi(\beta(\eta))}{Z(\eta)}, \quad (8)$$

$$\text{Var}[Y \mid X] = v(\eta, \sigma_\varepsilon) = \sigma_\varepsilon^2 \left[1 + \frac{\alpha(\eta) \varphi(\alpha(\eta)) - \beta(\eta) \varphi(\beta(\eta))}{Z(\eta)} - \left(\frac{\varphi(\alpha(\eta)) - \varphi(\beta(\eta))}{Z(\eta)} \right)^2 \right]. \quad (9)$$

The realized moments follow from Adam's and Eve's laws applied to the truncated conditionals. Since all parent values X_i are known at calibration time, we take expectations empirically, $\mathbb{E}[g(\eta, \sigma_\varepsilon)] = \frac{1}{n} \sum_{i=1}^n g(\eta_i, \sigma_\varepsilon)$ and $\text{Var}(g(\eta, \sigma_\varepsilon)) = \widehat{\text{Var}}(g(\eta_i, \sigma_\varepsilon)) = \frac{1}{n} \sum_{i=1}^n (g(\eta_i, \sigma_\varepsilon) - \bar{g})^2$:

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[m(\eta, \sigma_\varepsilon)] \quad [\text{Adam's law; def. of } m] \quad (10)$$

$$= \frac{1}{n} \sum_{i=1}^n m(\eta_i, \sigma_\varepsilon) \quad [\text{empirical convention}] \quad (11)$$

$$\text{Var}(Y) = \mathbb{E}[\text{Var}[Y \mid X]] + \text{Var}(\mathbb{E}[Y \mid X]) \quad [\text{Eve's law}] \quad (12)$$

$$= \mathbb{E}[v(\eta, \sigma_\varepsilon)] + \text{Var}(m(\eta, \sigma_\varepsilon)) \quad [\text{def. of } v, m] \quad (13)$$

$$= \frac{1}{n} \sum_{i=1}^n v(\eta_i, \sigma_\varepsilon) + \widehat{\text{Var}}(m(\eta_i, \sigma_\varepsilon)) \quad [\text{empirical convention}] \quad (14)$$

So, for known target moments μ_t and σ_t^2 , we set

$$\frac{1}{n} \sum_{i=1}^n m(\eta_i, \sigma_\varepsilon) = \mu_t \quad \text{and} \quad \frac{1}{n} \sum_{i=1}^n v(\eta_i, \sigma_\varepsilon) + \widehat{\text{Var}}(m(\eta_i, \sigma_\varepsilon)) = \sigma_t^2. \quad (15)$$

This leaves us with two equations in three unknown parameters $(\beta_0, \beta_1, \sigma_\varepsilon)$.

4 R^2 calibration (single parent)

For one-predictor OLS with an intercept, R^2 equals the squared sample correlation:

$$R^2 = \frac{\sum_i (\hat{Y}_i - \bar{Y})^2}{\sum_i (Y_i - \bar{Y})^2} \quad [\text{explained over total}]. \quad (16)$$

Since the OLS line passes through $(\bar{X}, \bar{Y})$, we have $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. So, substituting gives:

$$R^2 = \hat{\beta}_1^2 \frac{\sum_i (X_i - \bar{X})^2}{\sum_i (Y_i - \bar{Y})^2}. \quad (17)$$

Next, the OLS slope, $\hat{\beta}_1$ is defined by $\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}$. Squaring and substituting then gives:

$$R^2 = \frac{[\sum_i (X_i - \bar{X})(Y_i - \bar{Y})]^2}{[\sum_i (X_i - \bar{X})^2]^2} \cdot \frac{\sum_i (X_i - \bar{X})^2}{\sum_i (Y_i - \bar{Y})^2} \quad (18)$$

$$= \frac{[\sum_i (X_i - \bar{X})(Y_i - \bar{Y})]^2}{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2} = \hat{r}_{XY}^2 \quad [\text{def. of } \hat{r}_{XY}]. \quad (19)$$

Sample moments converge to population moments by the law of large numbers, and R^2 is a continuous function of them, so the continuous mapping theorem tells us R^2 approaches ρ_{XY}^2 as sample size increases:

$$R^2 \xrightarrow{p} \rho_{XY}^2 = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X) \text{Var}(Y)}. \quad (20)$$

Starting with the definition of covariance, $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$, we use Adam's Law,

$$E[XY] = E[E[XY | X]] \quad (21)$$

$$= E[X E[Y | X]] \quad [X \text{ constant when known}] \quad (22)$$

$$= E[X m(\eta, \sigma_\varepsilon)] \quad [\text{def. of } m] \quad (23)$$

and,

$$E[Y] = E[E[Y | X]] \quad (24)$$

$$= E[m(\eta, \sigma_\varepsilon)] \quad [\text{def. of } m]. \quad (25)$$

This gives $\text{Cov}(X, Y) = E[X m(\eta, \sigma_\varepsilon)] - E[X]E[m(\eta, \sigma_\varepsilon)] = \text{Cov}(X, m(\eta, \sigma_\varepsilon))$. So the observed covariance is the covariance of X with the conditional-mean curve.

From Section 3, $\text{Var}(Y) = E[v(\eta, \sigma_\varepsilon)] + \text{Var}(m(\eta, \sigma_\varepsilon))$.

Finally, substituting and applying the Section 3 empirical convention gives R^2 as a function of $(\beta_0, \beta_1, \sigma_\varepsilon)$:

$$R_{\text{map}}^2(\beta_0, \beta_1, \sigma_\varepsilon) = \frac{\widehat{\text{Cov}}(X_i, m(\eta_i, \sigma_\varepsilon))^2}{\widehat{\text{Var}}(X_i) \left[\frac{1}{n} \sum_{i=1}^n v(\eta_i, \sigma_\varepsilon) + \widehat{\text{Var}}(m(\eta_i, \sigma_\varepsilon)) \right]}, \quad (26)$$

where $\widehat{\text{Cov}}(X_i, m(\eta_i, \sigma_\varepsilon)) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(m(\eta_i, \sigma_\varepsilon) - \bar{m})$ is the empirical covariance. Since m and v are closed forms evaluated at the realized η_i , every quantity is computable at calibration time. Setting this equal to R_t^2 gives us our third equation.

This, combined with Section 3, leaves us with 3 equations (μ_t, σ_t^2 , and R_t^2) in 3 parameters $(\beta_0, \beta_1, \sigma_\varepsilon)$. This suggests we can solve directly for values that will allow us to at least approach our target R^2 .